Chapter-03: simplified of Boolean functions:

* **Ex-3.1:** obtain the simplified expressions in sum of products for the following Boolean functions.

(a) $F(x,y,z) = \Sigma(2,3,6,7)$

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Soln: we can simplified this expressions using K-map:

xyz	00	01	11	10
00			1	1
01			1	1

From the above table we find

$F = y.$

(b) $F(A,B,C,D) = \Sigma(7,13,14,15)$

Soln: using K-map we find:

AB \ CD	00	01	11	10
00				
01			1	
11		1	1	1
10				

from the above table we find,

$F_1 = ABD;\ F_2 = ABC,\ F_3 = BCD$

(94)

$\therefore F = F_1 + F_2 + F_3$

$= ABD + ABC + BCD$

(c) $F(A,B,C,D) = \Sigma(4,6,7,15)$

Soln: using k-map in this expression we find,

$F = BCD + A'BD'$.

(d) $F(w,x,y,z) = \Sigma(2,3,12,13,14,15)$.

Soln: using k-map in this expression we find.

$F = wx + w'x'y$

*Ex: 32: obtain the simplified expression in sum of products for the following Boolean functions:

(a) $xy + x'y'z' + x'yz'$

$= xyz + xyz' + x'y'z' + x'yz'$

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Now apply K-map in this expression we find,

x/yz	00	01	11	10
0	1			1
1			1	1

from the above table we find the simplified equation;

$$F = xy + x'z'$$

(b) $A'B + BC' + B'C'$

Soln: $F = A'BC + A'BC' + ABC' + A'BC' + AB'C' + A'B'C'$

$= A'BC + A'BC' + ABC' + AB'C' + A'B'C'$

now using K-map in this expressi[on] we find,

$$F = A'B + C'$$

(c) $a'b' + bc + a'bc'$

Soln: $F = a'b' + bc + a'bc'$

$= a'b'c + a'b'c' + abc + a'bc + a'bc'$

now, using K-map in this equation we find,

$F = a' + bc$

(d) $xy'z + xyz' + x'yz + xyz$

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Soln: $F = xy'z + xyz' + x'yz + xyz$

using K-map in this equation we find, $F = xy + xz + yz$.

* Ex-3.3: obtain the simplified expressions in sum of products for the following Boolean functions:

(a) $D(A'+B) + B'(C+AD)$

$= A'D(B+B') + BD(C+C') + B'C(A+A') + AB'D$

$= A'BD + A'B'D + BCD + BC'D + AB'C + A'B'C + AB'D$

$= A'BD(C+C') + A'B'D(C+C') + BCD(A+A') + BC'D(A+A') + AB'C(D+D') + A'B'C(D+D') + AB'D(C+C')$

$= A'BCD + A'BC'D + A'B'DC + A'B'C'D + ABCD + ABC'D + AB'CD + AB'CD' + AB'C'D.$

using k-map in this expression we find,

$$F = B'C + D$$

(b) $ABD + A'C'D' + A'B + A'CD' + AB'D'$

Soln: $F = ABD + A'C'D' + A'B + A'CD' + AB'D'$

followed by answer 3(a) we can simplified the expression using k-map. after using k-map we find:

$$F = BD + B'D' + A'B$$

(C) $k'lm' + k'm'n + klm'n' + lmn'$

Soln: $F = k'lm' + k'm'n + klm'n' + lmn'$

followed by answer 3(a) we can simplified this expression using k-map.

$$F = ln' + k'm'n$$

(d) $A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

Soln: $F = A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

Followed by answer 3(a) we use K-map in this expression and find the simplified expression,

$$F = B'D' + A'BD + ABC$$

(e) $x'z + w'xy' + w(x'y + xy')$

Soln: $F = x'z + w'xy' + wx'y + wxy'$

Followed by answer 3(a) we use K-map in this expression and find:

$$F = xy' + x'z + wx'y.$$

* Ex-3.4: obtain the simplified expressions in sum of products for the following Boolean functions:

(a) $F(A,B,C,D,E) = \Sigma(0,1,4,5,16,17,21,25,29)$

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Sol^n:

$\Rightarrow$ K-map for $A = 0$

BC/DE	00	01	11	10
00	1	1		
01	1	1		
11				
10				

$\Rightarrow$ K-map for $A = 1$

BC/DE	00	01	11	10
00	1	1		
01		1		
11		1		
10		1		

from the above table we find the simplified equations:

$$F = A'B'D' + B'C'D' + AD'E$$

(b) $BDE + B'C'D + CDE + A'B'CE + A'B'C + B'C'D'E$

Sol^n: Followed by answer 4(a) we can use K-map in this expression and find the simplified expression

$F = DE + A'B'C + B'C'E'$ (103)

(c) $A'B'CE' + A'B'C'D' + B'D'E' + B'CD' + CDE' + BDE'$

Soln: Followed by answer 4(a) we use K-map in this expression and find the simplified expression

$F = DE + A'B'C + B'C'E'$

*** Ex-3.5:** Given the following truth table:

x	y	z	F1	F2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(a) Express F_1 and F_2 in product of maxterms.

$F_1 = (x+y+z)(x+y'+z')(x'+y+z')(x'+y'+z)$

04

$= \Pi(0,3,5,6)$

$F_2 = (x+y+z)(x+y+z')(x+y'+z)(x'+y+z)$

$= \Pi(0,1,2,4)$

(b) obtain the simplified functions in sum of products

Soln:

$F_1 = x'y'z + x'yz' + xy'z' + xyz$

$= \Sigma(1,2,4,7)$

$F_2 = xy + xz + yz$

$= xy + xz + yz$.

(c) obtain the simplified functions in product of sums.

Soln: $F_1 = (x+y+z)(x+y'+z')(x'+y+z')(x'+y'+z)$

$F_2 = (x+y)(x+z)(y+z)$.

Ex-3.6: obtain the simplified expressions in product of sums. (05)

(a) $F(x,y,z) = \Pi(0,1,4,5)$

Solⁿ: For simplify the above pos expression we can use K-map:

x\yz	00	01	11	10
0	0	0		
1	0	0		

from above table we find the simplified equation is:

$F = y$

(b) $F(A,B,C,D) = \Pi(0,1,2,3,4,10,11)$

Solⁿ: to simplify the above pos expression we can use K-map:

AB\CD	00	01	11	10
00	0	0	0	0
01	0			
11				
10			0	0

from above table we find the simpliequation, $F = (B + C')(A + B)(A + C + D)$

(c) $F(w, x, y, z) = \prod(1, 3, 5, 7, 13, 15)$

Soln: For simplify the above pos expression we can use the k-map:

wx\yz	00	01	11	10
00		0	0	
01		0	0	
11		0	0	
10				

from the above table we find the simplified equation:

$$F = (x' + z')(w + z')$$

* Ex-3.7: obtain the simplified expression in (i) sum of products (sop) (ii) product of sums &

(a) $x'z' + yz' + yz' + xyz$

Soln: for simplify above expressirn we use k-map below: (107)

x/yz	00	01	11	10
0	1	0	0	1
1	1	0	1	1

from above table we find,

Sum of product (SOP) = $xy + z'$

product of sum (POS) = $(x+z')(y+z')$

(b) $(A+B'+D)(A'+B+D)(C+D)(C'+D')$

Soln: followed by answer 7(a) and using k-map we find the simplified expression:

$\Rightarrow$ SOP = $C'D + A'B'CD' + ABCD'$

$\Rightarrow$ POS = $(A+B'+D)(C+D)(C'+D')$

(c) $(A'+B'+D')(A'+B'+C')(A'+B+D')(B+C'+D')$

Soln: For simplifying Above expression we can use K-map;

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AB/CD	00	01	11	10
00	1	1	0	1
01	1	1	0	
11	1	0	0	1
10	1	0	0	1

from above table we find:

$SOP = F = B'D' + AD' + A'C'$

$POS = F = (A'+D')(B'+D')(A+B'+C')$

(d) $(A'+B'+D)(A'+D')(A+B+D')(A+B'+C+D)$

follow the solution 7(c) we can simplified the expression using K-map:

$SOP = B'D' + A'CD' + ABD$

$POS = (A'+B)(B+D')(B'+C+D)$.

(e) $w'yz' + uw'z' + uw'x + w'wz + uw'y'z'$

(109)

Sol^n:

For this expression we can find simplified expression using K-map. After using K-map we find:

$SOP = w'z' + uw'x' + u'wz$

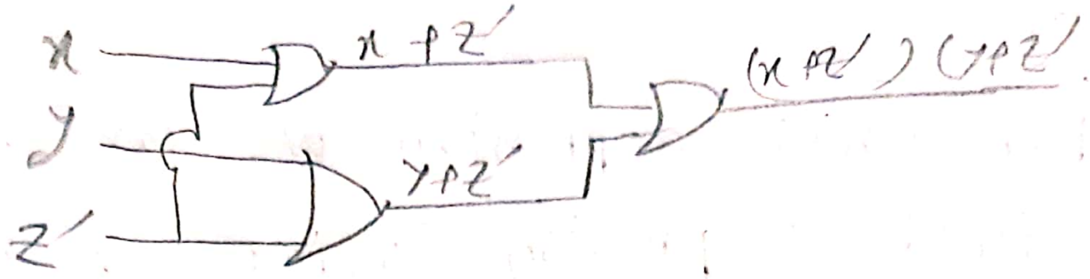
$POS = (u'+w')(w'+z)(w+x+z')(u+x+z')$

*** Ex-3.8:** Draw the gate Implementation of the simplified Boolean functions obtained in problem 1.7 using AND and OR gates.

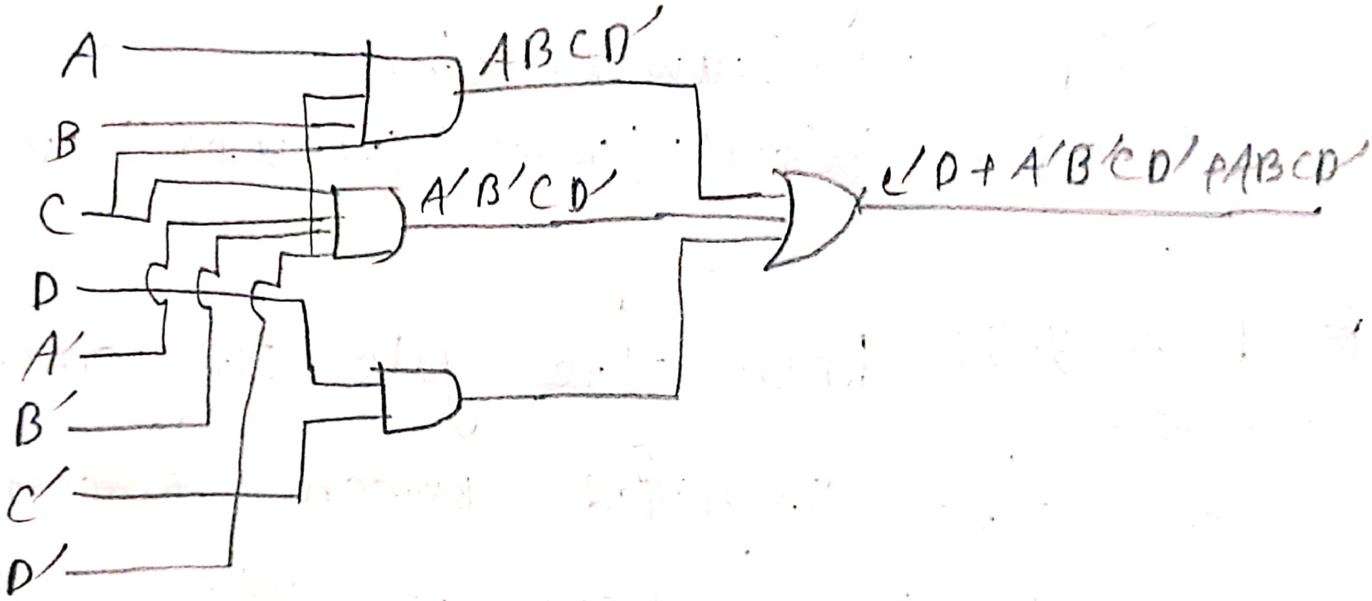
Sol^n:

(a) $SOP = xy + z'$

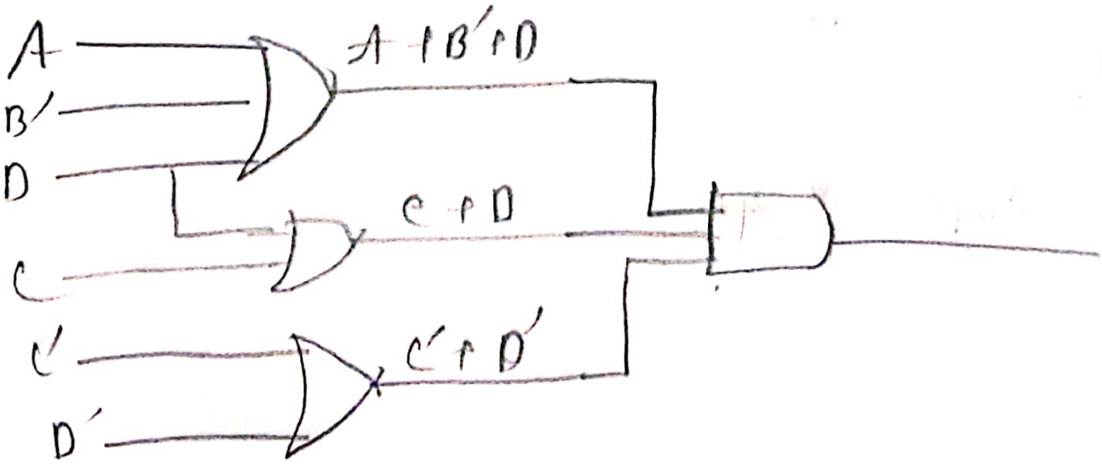
$\Rightarrow$ $POS = (x+z')(y+z')$



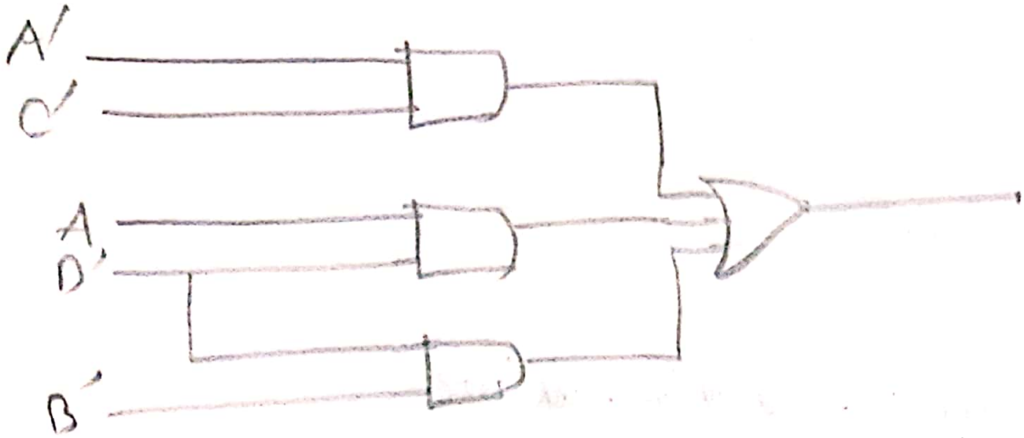
(b) $\Rightarrow$ $SOP = C'D + A'B'CD' + ABCD'$



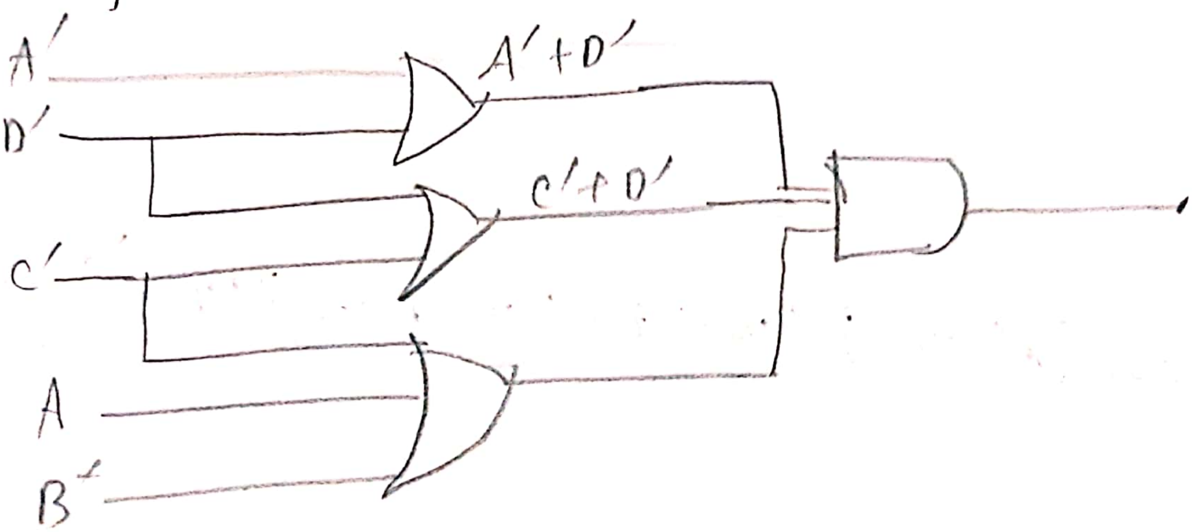
$\Rightarrow$ $POS = (A+B'+D)(C+D)(C'+D')$



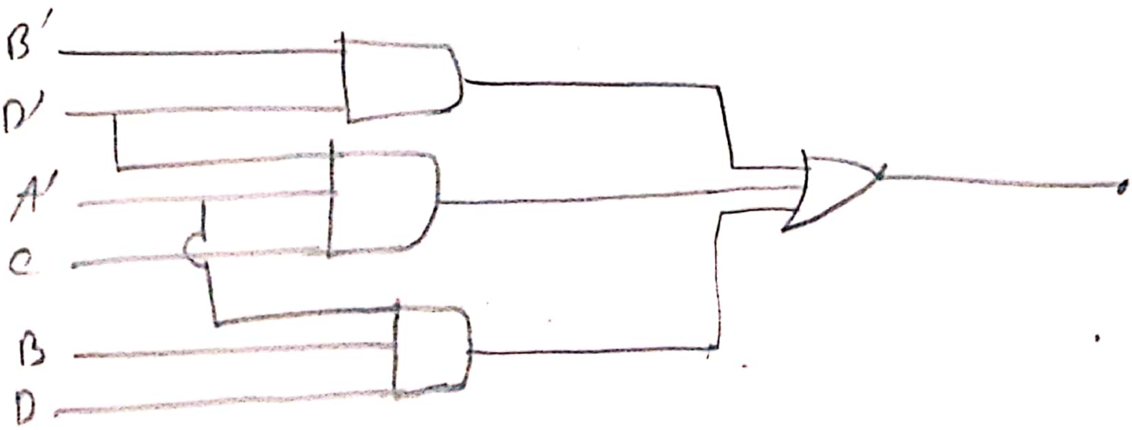
(c) $\Rightarrow$ SOP $= A'C' + AD' + B'D'$



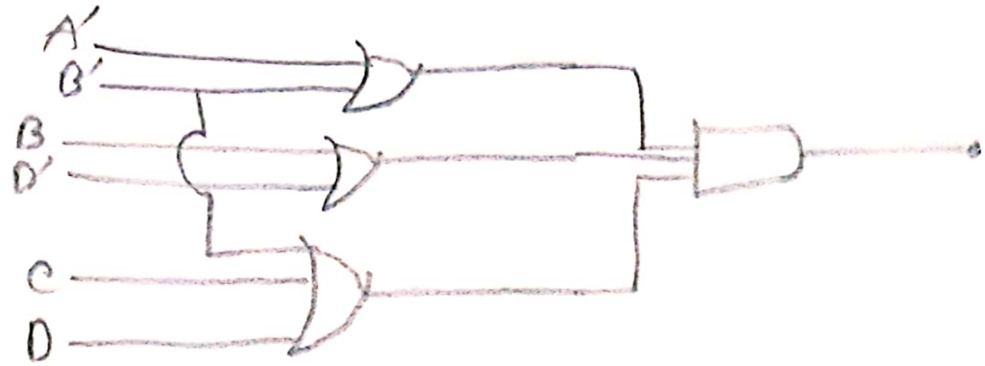
$\Rightarrow$ POS $:= (A'+D')(C'+D')(A+B'+C')$



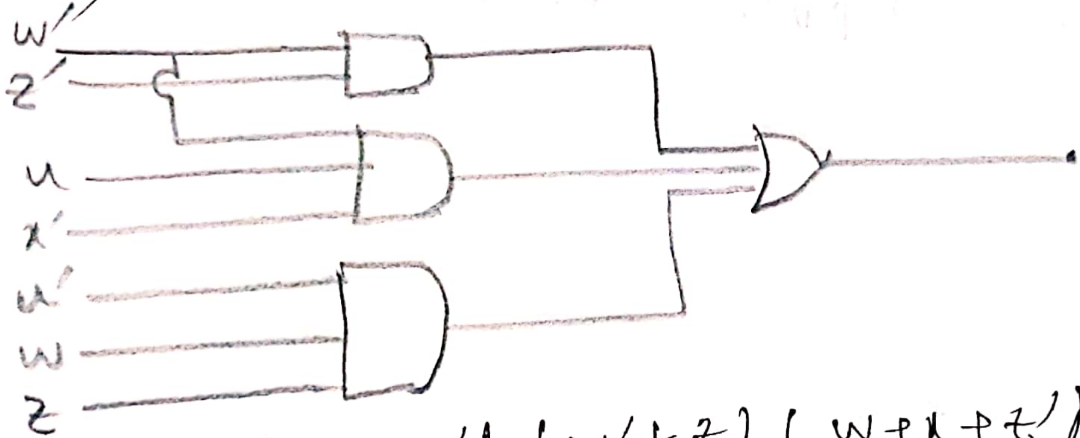
(d) SOP $= B'D' + A'CD' + A'BD$



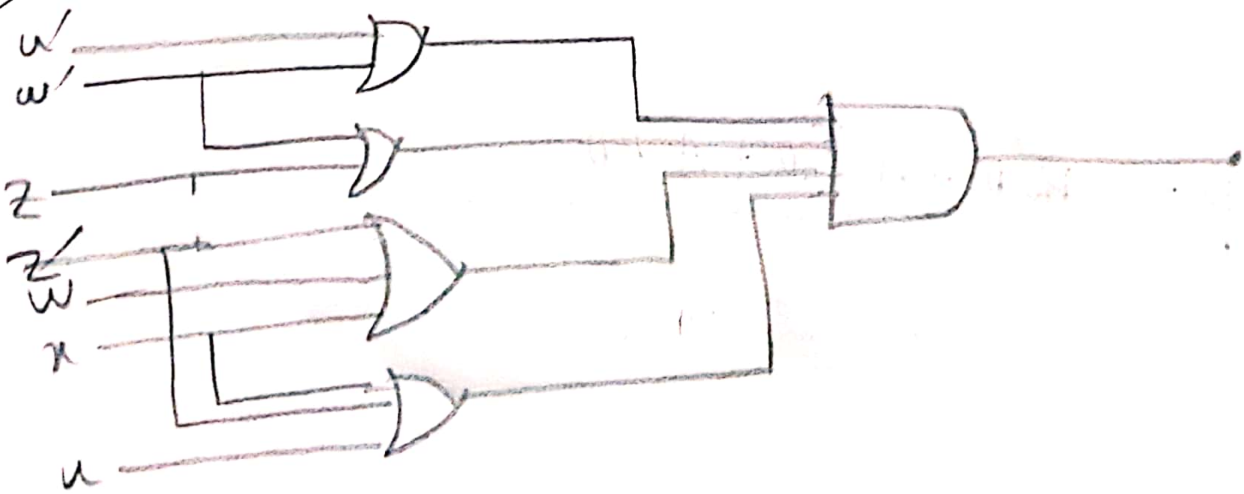
$\Rightarrow POS = (A'+B')(B+D')(B'+C+D)$



(e) $\Rightarrow SOP = w'z' + uw'x' + u'wz$



$\Rightarrow POS = (u'+w')(w'+z)(w+x+z')(u+x+z')$



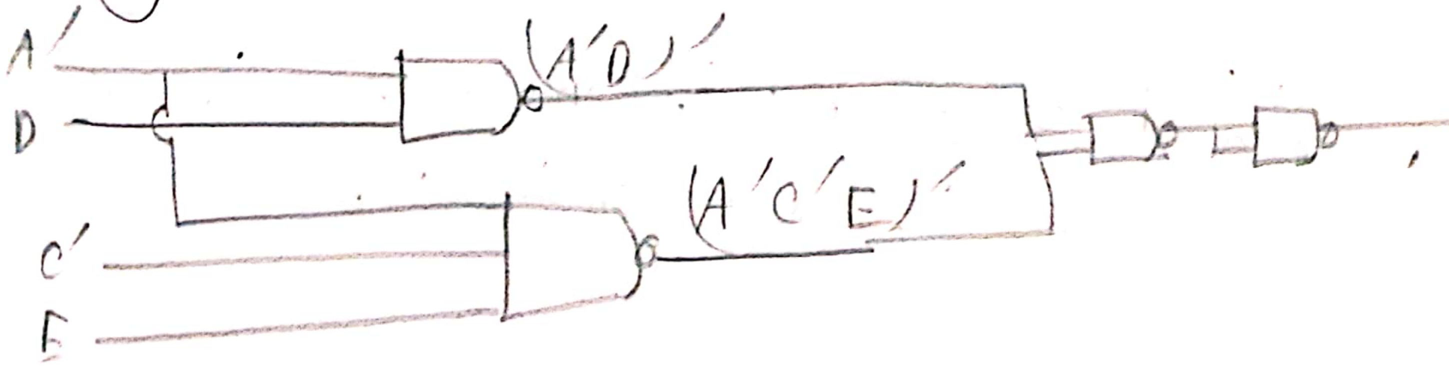
* E.X–3.9: Simplify each of the following expression and implement them using NAND gates. (113)

(a) $F_1 = AB' + AB'C + A'BD + ABE' + A'BE' + A'D$

Soln: Here, we use k-map and find the simplified expression:

$$F_1 = A + D'E' + CD'$$
$$= (A'D + A'C'E)'$$
$$= (A'D)'(A'C'E)'$$

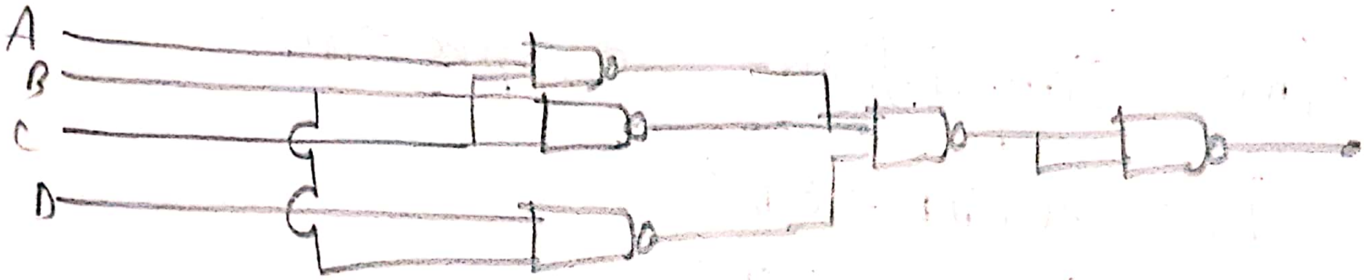
Now, we can draw the circuit diagram using NAND gate.



(b) $F_2 = (A+C)(A'+B'+C)(A'+B'+C+D)(B'+C+D')$

Here, we used K-map and find the simplified expression

$$F_2 = A'B' + C'D' + B'C'$$
$$= (BD + BC + AC)'$$
$$= (BD)' \cdot (BC)' \cdot (AC)'$$

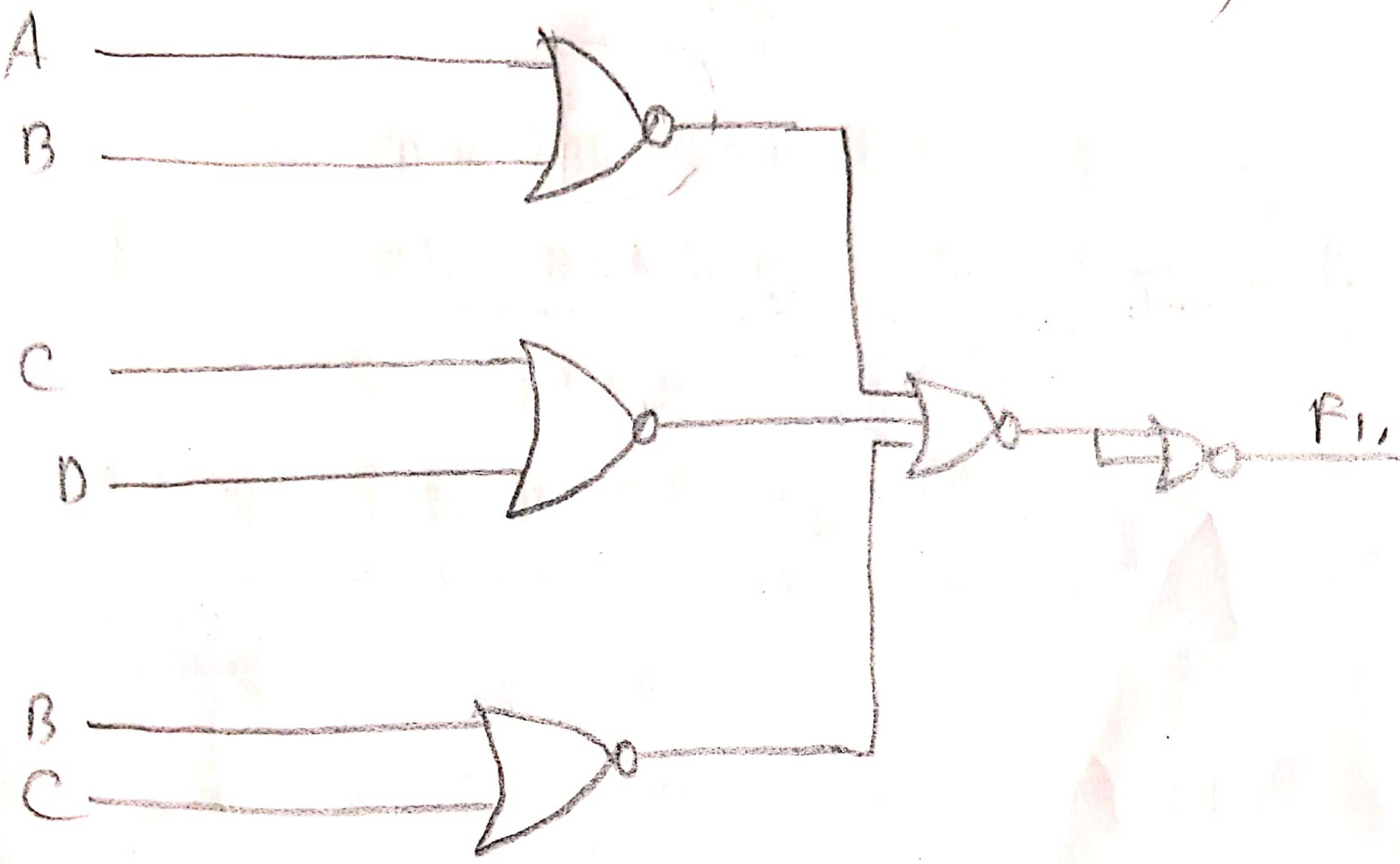


* **Ex-3.10:** Simplify each of the following functions and implement them with NOR gates.

(a) $F_1 = (A+B)(A'+B'+C+D')(A'+B'+D')$
$(A+B'+C+D')(C+D')$

⇒ After simplified. Now, we can impleme[nt] this using NOR gates.

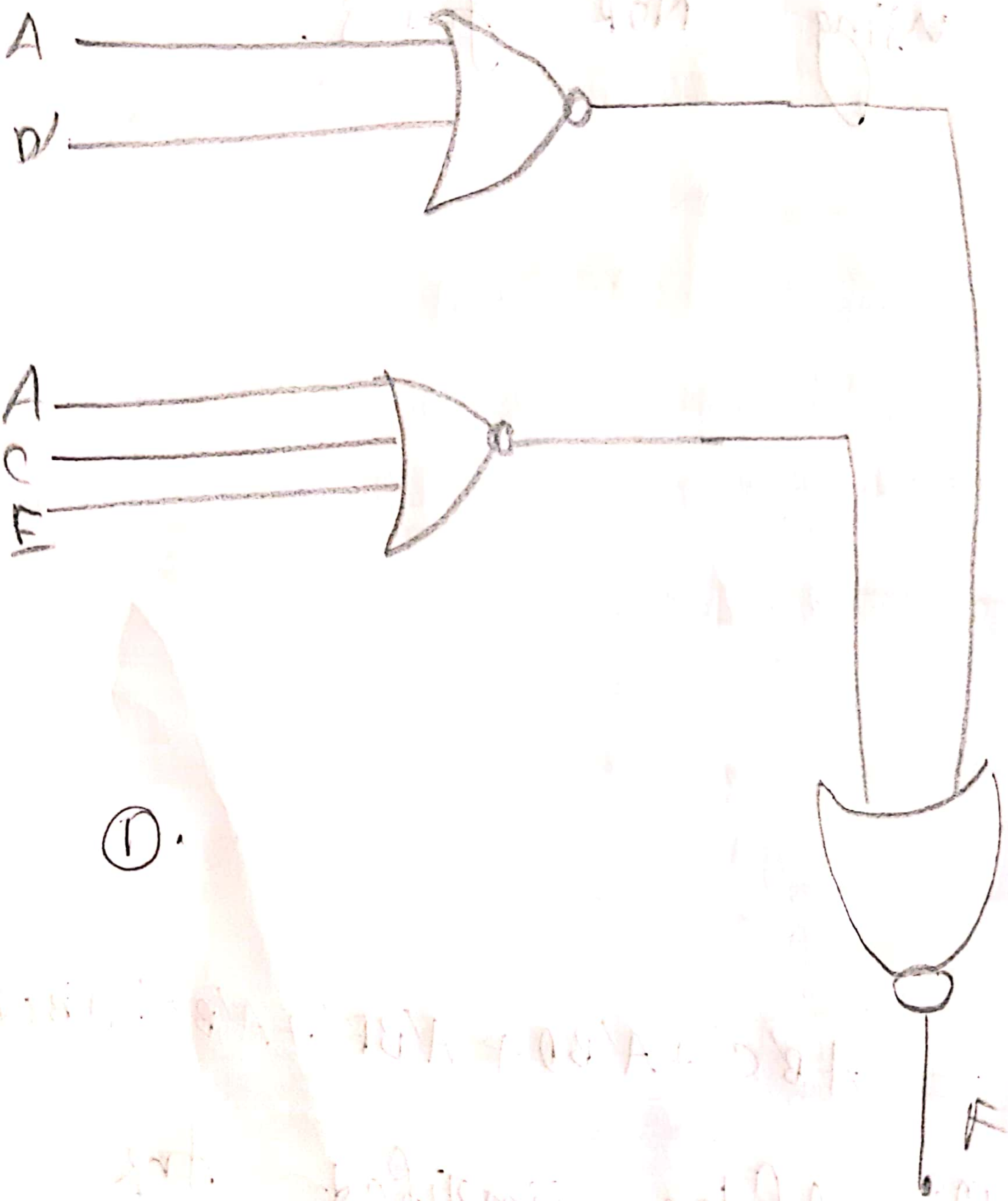
(115)



(b) $F_2 = AB'C + A'BD + A'BE' + A'BC' + A'BC'$+...

Soln: After simplified the expression. Now, we can implement it using NOR gates.

11.6



(I).

*E.x-3.11: Implement the following functions with NAND gates. Assume that both the normal and complement inputs are available.

(a) $BD + BCD + AB'CD' + A'B'CD'$

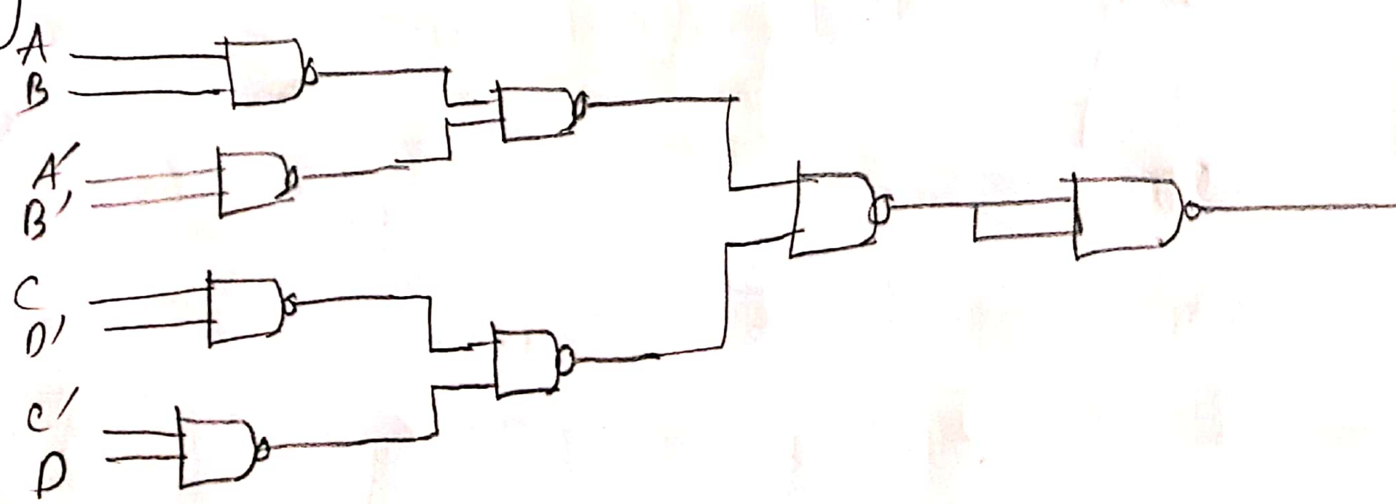
Soln: $F = BD + BCD + AB'CD' + A'B'CD'$

$= BD(C+1) + D'(AB'C + A'B'C)$

$= BD + D'(AB'C + A'B'C)$

now we can implement it using NAND gates.

(b) $(AB + A'B')(CD' + C'D)$ with two inputs gates:

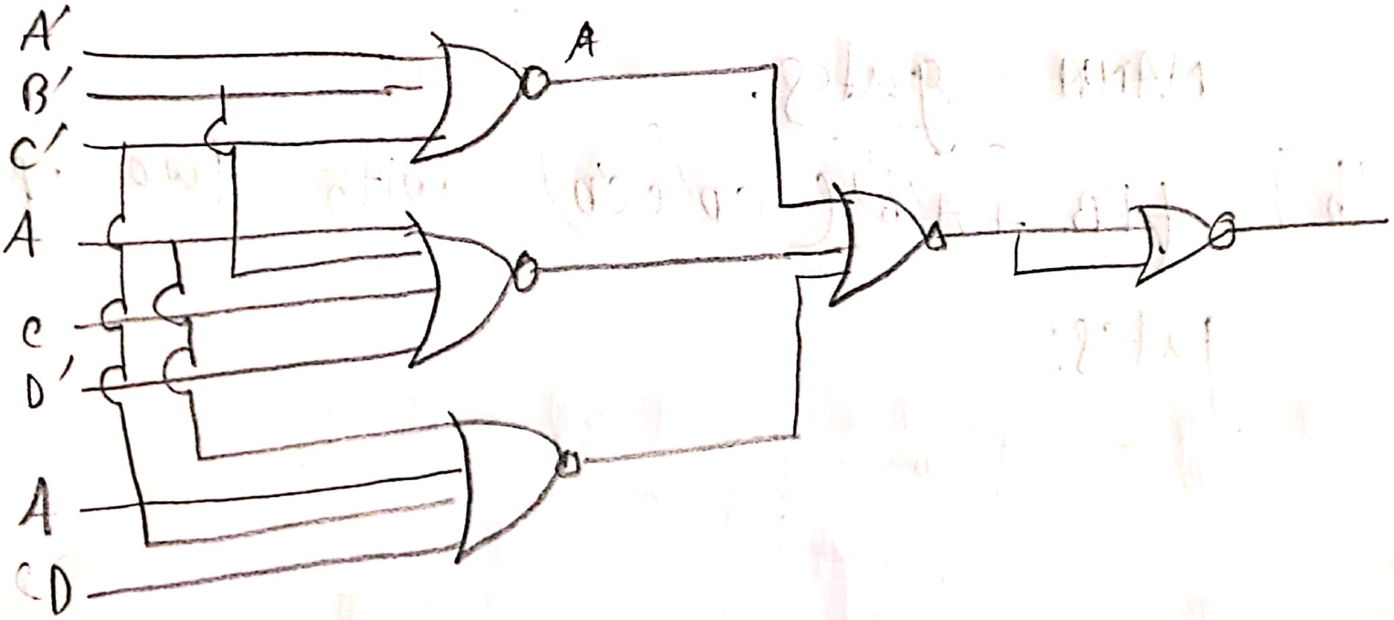


* **Ex-3.12 :** Implement the following functions with NOR gates. Assume that both the normal and complement inputs are available. (118)

(a) $AB' + C'D' + A'CD' + DC(AB + A'B') + DB(AC' + A'C)$

For simplifying above expression we use K-map and find:

$$F = (A' + B' + C')(A + B' + C + D')(A + B + C + D)$$

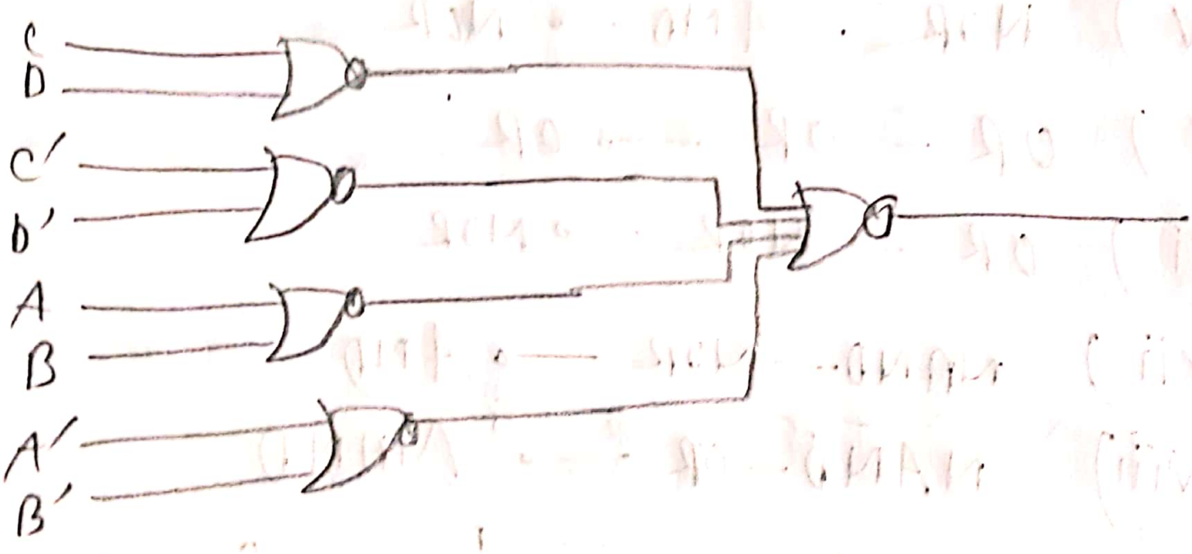


(b) $AB'CD' + A'BCD' + AB'C'D + A'BC'D$ (19)

Soln: for simplifying above expression we used K-map and find;

$$F = (C+D)(C'+D')(A+B)(A'+B')$$

Now, we can implement it using only NOR gates.



* Ex-3.13 List the eight degenerate two level forms and show that they reduce to a single operation. Explain how the degenerate two-level forms can be used to exten

the fan-in of gates.

Soln:

⇒ Eight degenerate two-level forms

(i) AND – AND → AND

(ii) AND – NAND → NAND

(iii) NOR – NAND → OR

(iv) NOR – AND → NOR

(v) OR – OR → OR

(vi) OR – NOR → NOR

(vii) NAND – NOR → AND

(viii) NAND – OR → NAND

⇒ used to extend the fan-in of gates: one application of de-generate two level forms is to extend the fan-in of gates. The fan-in of a gates is the number of inputs that it can accept and its typically limited by the physical characteristics of the gate itself. By using degenerate two-level forms, it is possible to combine multiple gates into a single gate with a higher fan-in which can reduce the overall complexity and improve the performance of a logic circuit.

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* **Ex-3.19:** Implement the functions of problem 3.9 with the following two level forms: NOR-OR, NAND-AND, OR-NAND, and AND-OR.

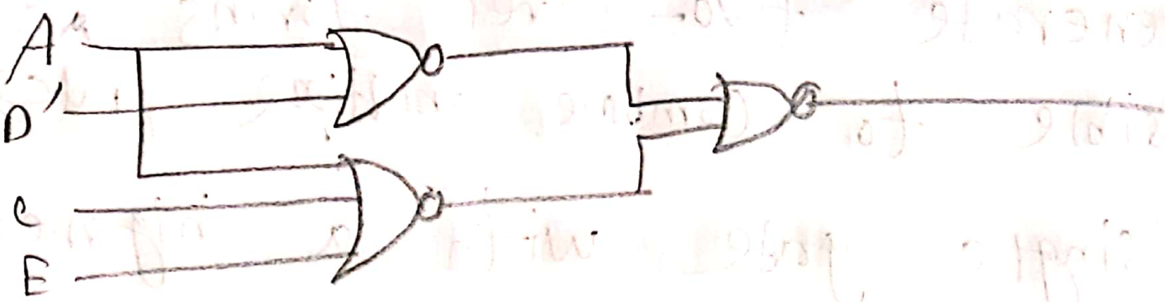
(a) $F_1 = A + D'B A C D'$

$= (A'D + A'C'B')'$

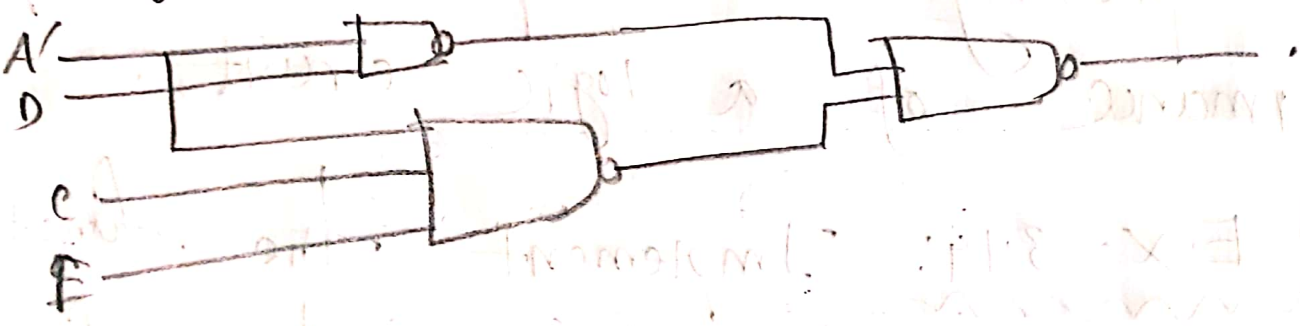
$= (A'D)'(A'C'B')'$

⇒ using NOR — OR:

$$F_1 = (A'D + A'C'E)'$$



⇒ using NAND - AND.

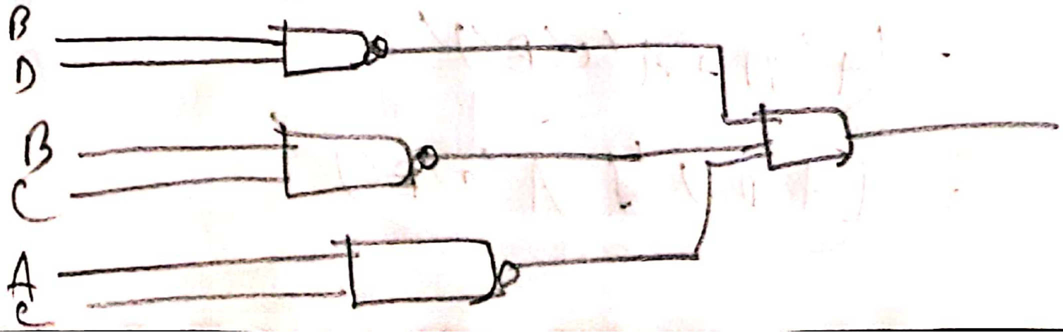


(b) $F_2 = A'B' + C'D' + B'C'$

$= BD + BC + AC$

$= ((BD)'(BC)'(AC)')'$

⇒ using NAND — AND



* Ex – 3.15: Simplify the following Boolean function F in sum of products using the don't care conditions d:

(a) $F = y' + x'z'$; $d = yz + xy$.

$F = xy'z + x'y'z' + xy'z' + xy'z' + x'yz'$

$d = xyz + xyz' + x'yz$

Now, we simplify this using K-map:

x/yz	00	01	11	10
0	1	1	x	1
1	1	1	x	x

The simplified form is $F = 1$

(b) $F = B'C'D' + BCD' + ABCD'$; $d = B'CD' + A'BC'D$

Soln: Similar to 15 (a) we used K-map and find:

$F = CD' + B'D' + ABCD$.

* Ex-3.16: Simplify the Boolean function F using don't care conditions d, in (1) sum of products and (2) product of sum:

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(a) $F = A'B'D' + A'CD + A'BC$

$= A'B'CD' + A'B'C'D' + A'BCD + A'B'CD + A'BCD'$

$d = A'BC'D + ACD + AB'D'$

$= A'BC'D + ABCD + AB'CD + AB'C'D' + AB'CD'$

using K-map we find

AB \ CD	00	01	11	10
00	1		1	1
01			1	1
11			X	
10	X		X	X

(1) $SOP = A'C + B'D'$

(2) $POS = A'(C+D')(B'+C)$.

(b) $F = w'(x'y + x'y' + xyz) + x'z'(y + w)$

$d = w'x(y'z + yz') + wyz$.

Similar to 16(a) we find

$SOP = x'z' + w'z$

$POS = (w' + z')(x' + z)$

(c) $F = ACE + A'CD'E' + A'C'DE$,

$d = DE' + A'D'E + AD'E'$

Soln:

Similar to 16(a) we find:

(i) $SOP = AC + CE' + A'C'D$

(ii) $POS = (A' + C)(C + E)(A + C + D)$

(d) $F = B'DE' + A'BE + B'C'E' + A'BC'D'$

$d = BDE' + CD'E'$.

Soln: similar to 16(a) we find:

(i) $SOP = A'B + B'E'$

(ii) $POS = (A' + B')(B + E')$.

* Ex-3.17: Implement the following functions using don't care conditions. Assume that both the normal and complement inputs are available.

(a) $F = A'B'C' + AB'D + A'B'CD'$ with no more than two NOR gates.

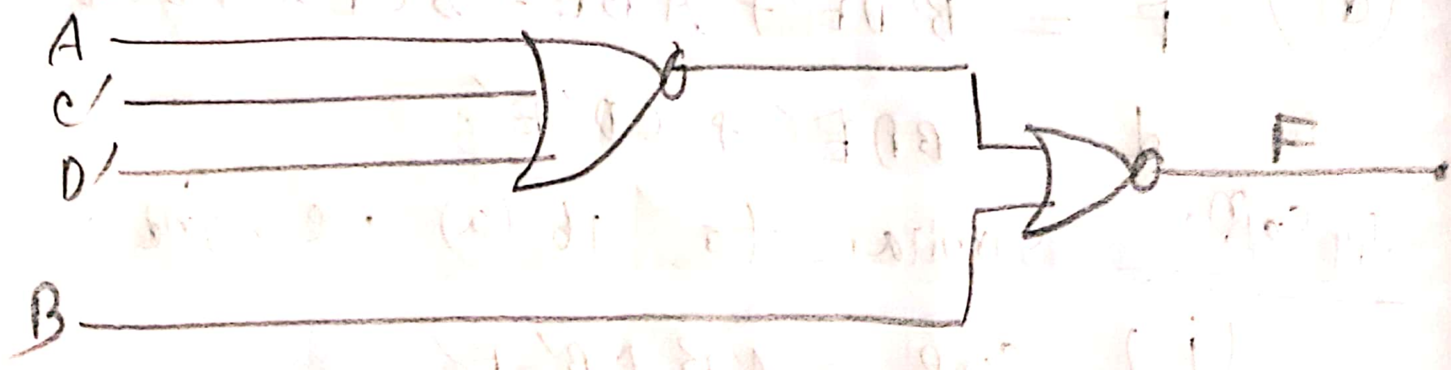
$d = ABC + AB'D'$

(126)

Soln: using K-map we find:

$$F = B'(A + C' + D')$$

Now, we can implement it using only two NOR gates.

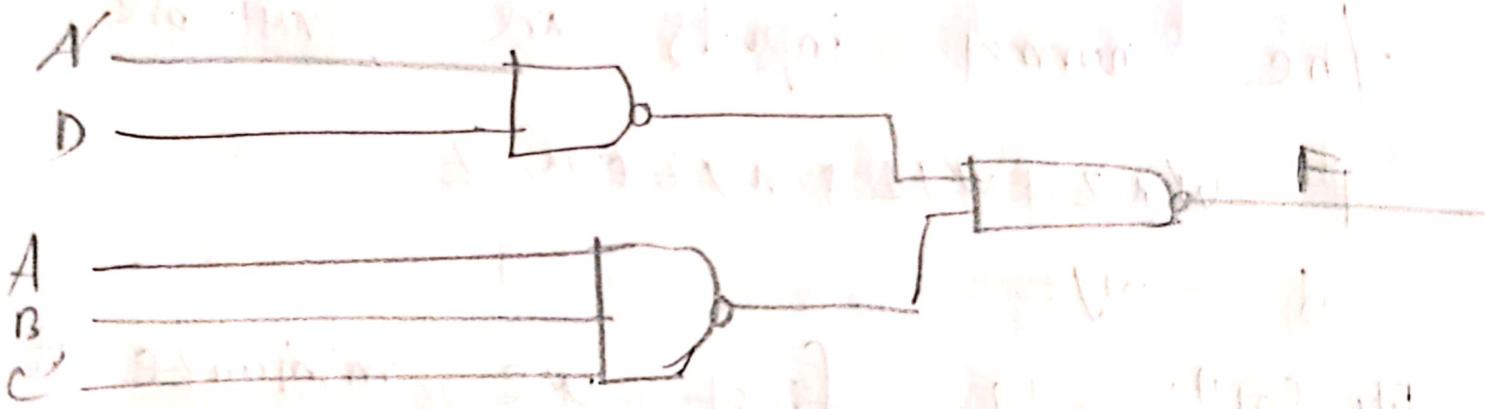


(b) $F = (A + D)(A' + B)(A' + C')$ with no more than three NAND gates. 127

Soln: For simplify we can use K-map and find:

$$F = A'D + ABC'$$

Now ~~eve~~ can implement it using ~~only~~ only NAND gates.



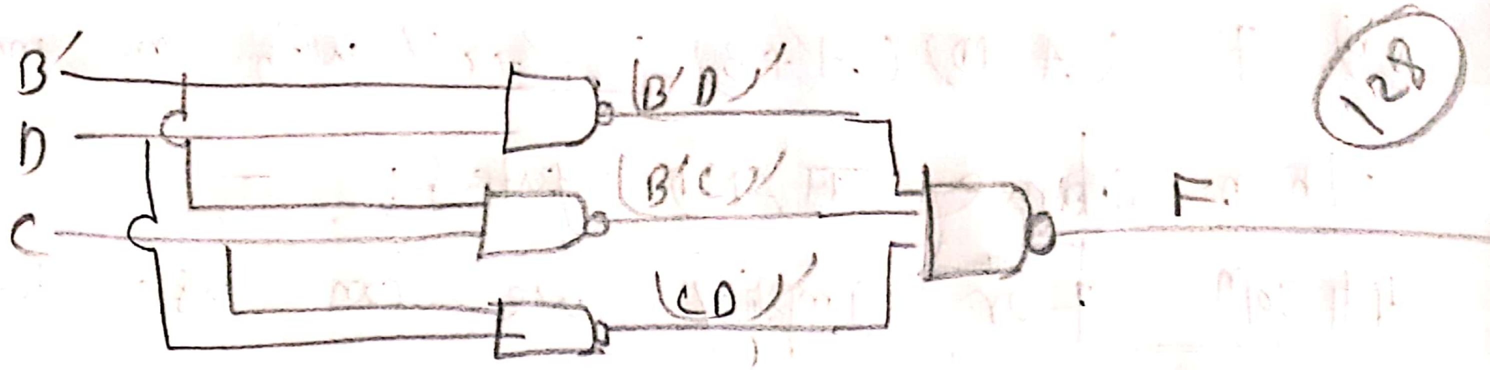
(c) $F = B'D + B'C + ABCD$ with NAND gates.

$$d = A'BD + AB'C'D'$$

Soln: for simplified we used K-map and find:

$$F = B'D + B'C + CD$$

Now, we can implement it using NAND gates



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*E.X-3.18: Implement the following function with either NAND or NOR gates. use only four gates. only the normal inputs are Available.

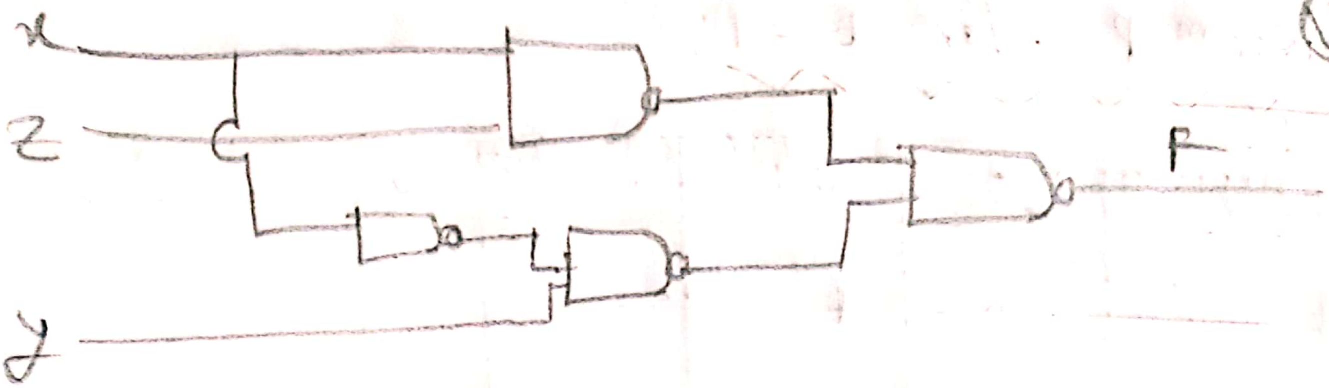
$F = w'xz + w'yz + x'yz' + wxy'z$

$d = wyz$

Soln: At first we minimize it using K-map and find :

$F = xz + x'y$

⇒ Now, we can implement it using NAND gates.



⇒ Similarly we can implement it using NOR gates.

*** Ex-3.19:** The following Boolean expression $BE + B'DE'$ is a simplified version of the expression: $A'BE + BCDE + BC'D'E + A'B'DE' + B'C'DE'$ Are there any don't care conditions? if so, what are they?

Soln:

⇒ For $E = 0$ k-map is:

AB / CD	00	01	11	10
00		1	1	
01				
11				
10		1	x	

⇒ K-map for E = 1:

AB/CD	00	01	11	10
00		1	1	
01	1	1	1	1
11	1	X	1	X
10				

So, the don't care conditions is:

$d(A, B, C, D) = (22, 27, 29)$.

* Ex - 3.20: Give three possible ways to express the function:

$F = A'B'D' + AB'CD' + A'BD + ABC'D$. with eight or less literals.

Soln: Here,

$F = A'B'D' + AB'CD' + A'BD + ABC'D$

$= B'D'(A' + AC) + BD(A' + AC')$

$= B'D'(A' + A)(A' + C) + BD(A' + A)(A' + C')$

$= B'D'(A' + C) + BD(A' + C')$ —— (1)

$= BB' + B'D'(A'+C) + BD(A'+C')$

$= B'[B+D'(A'+C)] + [B+D'(A'+C)][D(A'+C')]$

$= [B+D'(A'+C)][B'+D(A'+C')]$ —— (ii)

Again from (i), (131)

$F = BD' + B'D'\cdot(A'+C) + BD(A'+C')$

$= D'[D+B'(A'+C)] + [D+B'(A+C')$

$[B(A+C)]$

$= [D+B'(A'+C)][D'+B(A'+C')]$ —— (iii)

So, (i), (ii) and (iii) are required expressions.

*** Ex-3.21:** With the use of maps, find the simplest form in sum of products of the function $F = f \cdot g$, where f and g are given by:

$f = wxy' + y'z + w'yz' + x'yz'$

$g = (w+x+y'+z')(x'+y'+z)(w'+y+z')$.

Soln:

$$g' = w'x'yz + xyz' + wy'z$$

⇒ using K-map for f:

wx/yz	00	01	11	10
00		1		1
01	1	1		1
11		1		
10		1		1

⇒ using K-map for g:

wx/yz	00	01	11	10
00	1	1		1
01	1	1	1	
11	1		1	1
10	1		1	1

From the above table we find,

$$f \cdot g = x'yz' + w'xy'z + wxy'z'$$

* E.X-3.22: Simplify the boolean function of problem 3.2(a) using the map defined in fig-3.29(a). Repeat with the map of fig. 3.29(b). (133)

$F = xy + x'y'z' + x'yz'$

⇒ table for-a:

z\xy	00	01	11	10
0	0	2	6	4
1	1	3	7	5

⇒ table for-b:

xy\z	0	1
00	0	4
01	1	5
11	3	7
10	2	6

Soln: Here,

$$F = xy + x'y'z' + x'yz'$$
$$= xyz + xyz' + x'y'z' + x'yz'$$
$$= \Sigma(0, 2, 6, 7)$$

⇒ solve using table-a:

z \ xy	00	01	11	10
0	1	1	1	
1			1	

$$F = x'z' + xy.$$

⇒ solve using table-b:

zx \ y	0	1
00	1	
01		
11		1
10	1	1

$$F = x'z' + xy.$$

* E.X-3.23: simplify the Boolean function of problem 3.3(a) using the map defined in fig. 3.3(a). Repeat with the map of fig. 3.30(b).

(13.5)

$F = xy + x'y'z' + x'yz'$

⇒ table-a:

CD/AB	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

⇒ table-b:

BD/AC	00	01	11	10
00	12	14	6	4
01	13	15	7	5
11	9	11	3	11
10	8	10	2	0

Soln: Here,

$$F = A'D + BD + B'C + AB'D$$

$$= \Sigma(7, 5, 3, 1, 15, 13, 11, 10, 2, 1)$$

$$= \Sigma(1, 2, 3, 5, 7, 9, 10, 11, 12, 15)$$

now used k-map and find,

$$F = D + B'C$$

⇒ for table b:

AB\CD	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10		1	1	

$$F = D + B'C$$

Answer.

* **Ex-3.24:** Simplify the following Boolean functions by means of the tabulation method. 137

(a) $F(A,B,C,D,E,F,G) = \Sigma(20,28,52,60)$

Solⁿ: Convert the given minterms in their binary representation and arrange them according to number of one's present in the binary representation.

⇒ **Table-1:**

Group	minterms	A	B	C	D	E	F	G
0	20	0	0	1	0	1	0	0
1	28	0	0	1	1	1	0	0
	52	0	1	1	0	1	0	0
2	60	0	1	1	1	1	0	0

⇒ Table-2:

Group	pair	A	B	C	D	E	F	G
0	(20,28)	0	0	1	-	1	0	0
	(20,52)	0	-	1	0	1	0	0
0	(28,60)	0	-	1	1	1	0	0
	(52,60)	0	1	1	-	1	0	0

⇒ Table-3:

Group	Quad	A	B	C	D	E	F	G
0	(20,28,52,60)	0	-	1	-	1	0	0

$A'CEF'G'$ is obtained from table 3 as A, F, G contains '0' so $A'F'G'$, C, and E contains 1 so CE.

(b) $F(A, B, C, D, E, F, G) = \Sigma(20, 28, 38, 39, 52, 60, 102, 103, 127)$ 139

Soln: Similar to 2.24(a) we use tabular method from the above minterms expression and find,

$F = ABCDEFG + A'CEF'G' + BC'D'EF$

(c) $F(A, B, C, D, E, F) = \Sigma(6, 9, 13, 18, 19, 25, 27, 29, 41, 45, 57, 61)$

Soln: Similar to 2.24(a) we use tabular method from the above minterms expression and find,

$F = A'B'C'DEF' + A'BCD'E + CE'F + A'BD'EF$

* **Ex-3.25:** Repeat problem 3.6 using the tabulation method.

(a) $F = POS(0, 1, 4, 5)$

Soln:

⇒ table - a:

Group	x	y	z
0	0	0	0
1	0	0	1
4	1	0	0
5	1	0	1

⇒ table - b:

Group	x	y	z
(0,1)	0	0	-
(0,1)	-	0	0
(0,4)	-	0	0
(1,5)	-	0	1
(4,5)	1	0	-

⇒ table - c:

(0,1,4,5) = - 0 -

(0,1,4,5) = - 0 -

(0,1,4,5) = - 0 -

we find $F = y'$.

(b) $F = pos(0,1,2,3,9,10,11)$

Soln: similar to 25(b) we simplified it using tabular method and find.

$$F = (B+C')(A+B)(A+C+D)$$

(c) $F = pos(1,3,5,7,13,15)$

Soln: similar to 25(c) we simplified it using tabular method and find:

$$F = (w+y')(x'+z')$$

* E.X-3.26: Repeat problem 3.16 (c) and (d) using the tabulation method.

(c) $F = ACE + A'CD'E' + A'C'DE$

$d = DE' + A'DE + AD'E'$

Soln: Here,

$$F = ACE + A'CD'E' + A'C'DE$$

$$= ACDE + ACD'E + A'CD'E' + A'C'DE.$$

$= \Sigma(3, 4, 13, 15)$

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and $d = DE' + A'D'E + AD'E'$

$= ACDE' + AC'DE' + A'CDE' + A'C'DE' + A'CD'E$
$+ A'C'D'E + ACD'E' + AC'D'E'$

$= \Sigma(14, 10, 6, 2, 5, 1, 12, 8)$

$= \Sigma(1, 2, 5, 6, 8, 10, 12, 14)$

now, we can use tabular method and find,

$F = A'C'E + CD' + AC.$

(d) $F = B'DE' + A'BE + B'C'E' + A'BC'D'$

$d = BDE' + CD'E'$

Soln: $F = B'DE' + A'BE + B'C'E' + A'BC'D'$
$+ AB'C'DE + A'B'CDE' + A'B'C'DE'$

$= AB'CDE' + AB'C'DE + A'B'CDE' + A'BC'D'E$
$+ A'BCDE + A'BCD'E + A'BC'DE + A'BC'D'E$
$+ AB'C'DE' + AB'C'D'E' + A'B'C'DE' + ABCDE$
$+ A'BC'DE + A'BCD'E$

$= \Sigma(0, 2, 6, 8, 9, 11, 13, 15, 16, 18, 22)$

$$d = ABCDE' + ABC'DE' + A'BCDE' + A'BC'DE' + ABCD'E'$$
$$+ AB'CD'E' + A'BCD'E' + A'B'CD'E'$$

(143)

$$= \Sigma(4, 10, 12, 14, 20, 26, 28, 30)$$

now, we can simplified these using tabulation method and find:

$$F = B'E' + A'B$$

*** Ex-3.27:** simplify $f(w, x, y, z) = \Sigma(0, 1, 2, 8, 12, 13, 14) + d(3, 5, 10, 15)$ with K-map and implement using two variable NAND gates.

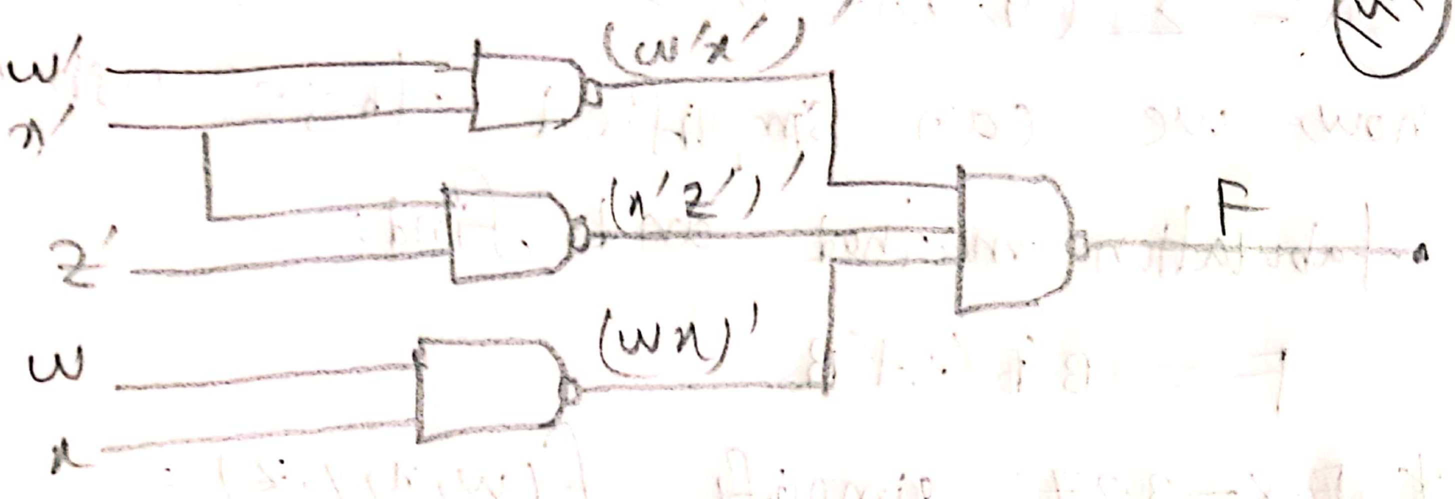
Soln:

wx/yz	00	01	11	10
00	1	1	×	1
01	×			
11	1	1	×	1
10	1			×

After simplified we find, $F = w'x' + wx + y'z'$

⇒ Now, we can implement this using two variable NAND gates.

(144)



* **Ex - 3.28:** What are prime implicants? Minimize following equation using Quine-McCluskey method.

$$f(v,w,x,y,z) = \Sigma(1,4,12,14,16,18,21,25,26,29,31) + d(0,2,5,30).$$

Sol^n:

⇒ prime implicants: A group of squares or rectangles made up of a bunch of adjacent minterms.

that is allowed by definition of a k-map are known as prime implicants or PI, i.e. all the possible groups that are formed in k-map. (145)

⇒ count the number of 1's in each min term given in the problem.

minterms	Binary	No. of 1's
0	00000	0
1	00001	1
2	00010	1
4	00100	1
5	00101	2
12	01100	2
14	01110	3
16	10000	1
18	10010	2
21	10101	3
25	11001	3
26	11010	3
29	11101	4
30	11110	4
31	11111	5

⇒ Now, we can grapp them and according to the base of 1's and finay we can find the simplified expression.